PATENT SEARCH REPORT

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Ref No: BMU 38

Suggested Title: METHOD FOR PROCESSING IMAGE CAPTURED THROUGH CAMERA

Abstract / Scope: The present invention relates to field of security systems. More specifically, the present invention relates to a method for processing image through different module such as; multithreading module, image detection module, feature extraction module and training module.

Key features:

The present invention relates to smart surveillance camera. More particularly, the present invention relates to a method for recognizing faces from the images captured by the camera by comparing them with the images stored in a database.

The method comprises the following steps;

1. Capturing image using a camera.
2. Sending the images on server storage using a microcontroller.
3. Polling the images using a multithreading module.
4. Detecting faces from the images through Haar cascade, HOG(Histogram of oriented gradients) and Linear SVM(Support vector Machine)
5. Computing face embeddings of the extracted features using a deep metric network.
6. Comparing image to predefined image stored on the server.

Keywords:

Keywords	Synonyms
Camera	Capture Image
Surveillance	Observation, View, Watch, Supervision
Microcontroller	Control Unit, Processor
Image	Picture, Figure, Photo
Detect	Recognize, Observe, Identify

Database Search (Tools)

Orbit, Patent Lens, Free Patents Online, USPTO, Espace Net

Non-Patent Literature Search (Tools)

Science Direct, Google Scholar

SCOREEVALUATION:

The report has a scoring system on a scale of 5 i.e.

Score of 1 - Completely irrelevant

Score of 2 - Too an extent relevant

Score of 3 - Relevant

Score of 4 - Closely related

Score of 5 - Completely relevant

Prior Art

Patent Number	KR101793510B1
Title	System for learning and recognizing face and thereof method
Priority date	2017-03-31
Inventor	Lee Seung Ho, Jang Kyung Wook
Assignee	Hanbat National University
Abstract	The present invention presents a facial recognition system to learn and how. And IP camera unit to face learning system is multi-angle receive a face-up image in accordance with the present invention, learning the face extraction in the face detection and the face detection unit for extracting a learning target face in the multi-layered face-up image using a deep learning techniques which includes parts of the face learning. According to the invention, the progress of learning target face learning by using the face-up image of the face multiple angles have been recorded with multiple angles, so it is possible to identify facial features of each angle to study the face, in addition learning the face on the basis of deep learning techniques so it can increase the face learning precision and reliability, the also multi-angle face shot image and a recognition target face compare to learning the traditional face and awareness can be determined that the target faces are matched, the recognition target face using deep learning techniques since classification can increase the face recognition.
Close Features To Present Invention	KR' 510 relates to face learning and recognition system and method and, more particularly, to using multiple cameras shooting the recognition target face from various angles and to analyze, classify

	and study the faces photographed using deep learning skills It relates to facial-recognition technology. 1. Discloses about haar cascade, SVM for image processing. 2. Discloses about controller for data processing. 3. Discloses about storage unit for storing the data. 4. Discloses about extracting image through deep learning. 5. Discloses about comparing and analyzing image to predefined image stored on the storage unit.
Feature Not Present	1. Does not disclose about HOG image processing. 2. Does not disclose about polling frames using multithreading. 3. Does not explicitly talk about server storage.
Score	**3 OUT OF 5**

Patent Number	WO2017197620A1
Title	Detection of humans in images using depth information
Priority date	2016-05-19
Inventor	Haibing Ren, Yimin Zhang, Fei Duan
Assignee	Intel Corporation
Abstract	Techniques are provided for detection of humans in images that include depth information. A methodology embodying the techniques includes segmenting an image into multiple windows and estimating the distance to a subject in each window based on depth pixel values in that window, and filtering to reject windows with sizes that are outside of a desired window size range. The desired window size range is based on the estimated subject distance and the focal length of the depth camera that produced the image. The method further includes generating classifier features for each remaining windows (post-filtering) for use by a cascade classifier. The cascade classifier creates candidate windows for further consideration based on a preliminary detection of a human in any of the remaining windows. The method further includes merging neighboring candidate windows and executing a linear classifier on the merged candidate windows to verify the detection of a human.
Close Features To Present Invention	WO' 620 relates to context surrounding the candidate windows. 1. Discloses about haar cascade, HOG and SVM for image processing. 2. Discloses about processor for data processing. 3. Discloses about server storage unit for storing the data. 4. Discloses about comparing and analyzing image to predefined image stored on the storage unit.

Feature Not Present	1. Does not disclose about polling frames using multithreading. 2. Does not disclose about deep metric network for image extraction.
Score	**3 OUT OF 5**

Patent Number	CN105335722A
Title	Detection system and detection method based on depth image information
Priority date	2015-10-30
Inventor	Zhang Wei, Wu Zihao, Tang Xiaoou
Assignee	Shangtang Group Co., Ltd.
Abstract	Embodiments of the invention disclose a detection system and a detection method based on depth image information. The system comprises a camera device used for acquiring a depth image of an object to be detected, an extraction device used for extracting a target area of the object to be detected based on the depth image and acquiring the feature point information of the target area, and a detection device used for detecting whether the object to be detected is a living body based on the target area and the feature point information. Through face identification based on the scheme provided by the embodiments of the invention, others can be prevented from pretending to be someone with a photo, a video or other nonliving object to pass adult face identification.
Close Features To Present Invention	CN' 722 relates to an image recognition technique, and more particularly relates to detection systems and methods based on the depth image information. 1. Discloses about haar cascade, HOG and SVM for image processing. 2. Discloses about multithreading for image extraction. 3. Discloses about storage medium for storing the data. 4. Discloses about comparing and analyzing image to predefined image stored on the storage unit.

Feature Not Present	1. Does not disclose about polling frames using multithreading. 2. Does not disclose about deep metric network for image extraction. 3. Does not disclose about polling frames using multithreading.
Score	**3 OUT OF 5**

Patent Number	US20120274755A1
Title	System and method for human detection and counting using background modeling, hog and haar features
Priority date	2011-06-15
Inventor	Aniruddha Sinha, Rohit Gupta, Chaki Chain, Arpan Pal
Assignee	Tata Consultancy Services Ltd
Abstract	A system for adaptive learning based human detection for channel input of captured human image signals, the system comprising: a sensor for tracking real-time images of an environment of interest; a feature extraction and classifiers generation processor for extracting a plurality of features and classifying the features associated with time-space descriptors of image comprising background modeling, Histogram of Oriented Gradients (HOG) and Haar like wavelet; a processor configured to process extracted feature classifiers associated with plurality of real-time images; combine the plurality of feature classifiers of time-space descriptors; evaluate a linear probability of human detection based on a predetermined threshold value of the feature classifiers in a time window having at least one image frame; a counter for counting the number of humans in the real-time images; and a transmission device configured to send the final human detection decision and number thereof to a storage device.
Close Features To Present Invention	US' 755 relates to a system and method for image information processing, and more particularly relates to a precise human detection by distinguishing the target object from background by combining different algorithms. 1. Discloses about haar cascade and HOG image processing.

	2. Discloses about microcontroller for data processing. 3. Discloses about server storage for storing data.
Feature Not Present	1. Does not disclose about polling frames using multithreading. 2. Does not disclose about SVM for image processing. 3. Does not disclose about deep metric network for image extraction.
Score	**2 OUT OF 5**

Patent Number	US20180330171A1
Title	Methods for detecting, identifying and displaying object information with a multi-camera vision system
Priority date	2017-05-10
Inventor	Peter Corcoran
Assignee	Fotonation Ltd
Abstract	Methods for detecting, identifying and displaying object information with a multi-camera vision system. In one embodiment presence of an object is detected in a region extending above a surface and positioned beyond a border, the region including at least first and second zones. A frame of image data is acquired which spans a field of view angle corresponding to a portion of a region from a position along the border. Object detection criteria are applied to analyze and classify an image of an object among a set of object types. A determination is made whether the object is present in the first zone based on application of a first series of scan windows varying in size over a first range of window sizes relative to the size of the image frame, this only permitting detection of images limited in size to a first range of image sizes.
Close Features To Present Invention	US' 171 relates to systems and methods for monitoring activity in a region surrounding a boundary or border. Embodiments of the invention provide real-time awareness of presence about a structure. In one example, a vision system monitors and displays roadway traffic activities about a moving vehicle. 1. Discloses about haar cascade, HOG for image processing. 2. Discloses about CAN controller for data processing.

	3. Discloses about data storage unit for storing the data. 4. Discloses about analyzing image to predefined image stored on the storage unit.
Feature Not Present	1. Does not disclose about SVM image processing. 2. Does not disclose about polling frames using multithreading. 3. Does not explicitly talk about server storage. 4. Does not disclose about deep metric network for image extraction.
Score	**2 OUT OF 5**

2. Non-Patented Literature

NPL	LINK
Title	Face Recognition Based on HOG and Fast PCA Algorithm
Publication Year	2018
Author	Xiang-Yu Li, Zhen-Xian Lin
Comment	A new method of face recognition based on gradient direction histogram (HOG) features extraction and fast principal component analysis (PCA) algorithm is proposed to solve the problem of low accuracy of face recognition under non-restrictive conditions. In this method, the Haar feature classifier is used to extract and extract the original data, and then the HOG features are extracted from the image data and the PCA dimension reduction is processed, and the Support Vector Machines (SVM) algorithm is used to recognize the face. The experimental results of the classification recognition on the LFW face database verify the effectiveness of the method.

Comments

According to the disclosure the major objectives of the work were studied for the invention and scope of search was thoroughly reviewed.

The aforementioned prior art **KR101793510B1** and **NPL titled "Face Recognition Based on HOG and Fast PCA Algorithm"** discloses the processing of images through different modules for enhancing the quality of images captured by a camera. The cited documents discloses some of the prominent aspects of the present invention like Haar cascade, HOG (Histogram of Oriented Gradients) and SVM (Support Vector Machine) for image processing, a controller for processing data, a storage unit for storing data, extracting image through deep learning.

But the cited documents does not disclose about the polling of images through multithreading and feature extraction of images using a deep metric network.

Hence, the present invention has novelty feature(s). So, we recommend inventor(s) to proceed for the further process.

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